

# **Where Occupations Move: Interstate Migration by Occupation in the United States, 2005 to 2024**

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## **Extended summary**

Interstate migration in the United States has been measured extensively for persons and rarely for occupations, which leaves a gap between the evidence that exists and the questions that states put to it, since most of the rules that make moving difficult attach to jobs, and since a state deciding whether to train workers or to recruit them needs to know what share of its workforce in a given occupation arrives from outside and who those arrivals are.

This paper builds bilateral interstate migration flows by occupation from nineteen vintages of American Community Survey microdata covering 2005 to 2024, 59.8 million person records across 51 jurisdictions and 479 harmonised occupation codes, and uses them to measure the share of each state's occupational workforce that arrived from another state within the year. That share ranges thirteenfold across occupations, from 0.66 per cent for tool and die makers to 9.70 per cent for sailors, and the design-based confidence intervals at the two ends of the distribution do not overlap.

Three results follow from the panel. Interstate migration delivers young workers and delivers them symmetrically: arrivals average 34.0 years of age, leavers average 34.0, the workforce they join or leave averages 41.3, and in every occupation with enough arrivals to measure the incomers are younger than the people already doing the job. A gravity model estimated by Poisson pseudo-maximum likelihood puts the elasticity of flows with respect to the origin-destination wage differential at 0.114 against a distance elasticity of roughly  $-0.69$ , so that a ten per cent pay advantage buys about one per cent more migration while distance carries some six times the weight. Deviations from the gravity benchmark reproduce occupational concentrations that were never supplied to the model, placing Washington state at 2.82 times its predicted intake of software developers, the District of Columbia at 3.64 times its predicted intake of lawyers, and Michigan at 4.16 times its predicted intake of mechanical engineers.

The measures are tested against the Occupational Employment and Wage Statistics, an establishment survey collected independently of the household microdata from which every migration figure here is built. Ten hypotheses were specified before the results were examined and all ten are reported, with  $p$  values corrected across the battery. Three survive correction: the gravity residual tracks the location quotient, external dependence predicts employment growth, and net inflow predicts employment growth. The three wage tests return coefficients of  $-0.0004$ ,  $-0.0014$  and  $+0.0001$  against standard errors between 0.004 and 0.007, which is precise enough to exclude a wage relationship of any size a policy could act on. Two unrelated collections therefore place interstate migration where employment is expanding, and neither finds a measurable connection to

where pay is rising.

## 1 Introduction

Roughly 2.3 per cent of employed Americans crossed a state line in a given year over the period studied here, and the stability of that figure across two decades conceals a distribution that is wide throughout. Tool and die makers replenish two thirds of one per cent of their number from other states each year while sailors replenish close to a tenth; farmers, postal workers, bus drivers and highway maintenance workers sit within a percentage point of the floor, and pilots, air traffic controllers, aircraft mechanics and physical scientists sit five to nine times above it. A policy instrument aimed at labour mobility acts somewhere inside that distribution, which has not previously been measured at occupation detail across a panel of this length.

The absence follows from what the available data could support. The migration literature has treated occupation coarsely or not at all, from Molloy et al. (2011) on the aggregate decline to Kennan and Walker (2011) on the individual decision, while the literature that does resolve occupation, chiefly the studies of individual licensure compacts, treats one occupation at a time. This paper constructs the intermediate object: bilateral flows by occupation, state and year, over a period long enough to observe change in them.

The question a workforce authority puts to such data is a narrow one. It asks whether the people filling a given occupation locally were trained locally or arrived from elsewhere, whether the arrivals are the ones needed, and whether recruitment or training is the instrument more likely to work, and all three require bilateral flows resolved by occupation and by age.

The claims made here are descriptive, and no causal effect of any policy is identified. The contribution is a set of measures constructed from public data, reported with the uncertainty the survey design implies, and tested against a source from which they were not built.

Section 2 describes the data and the construction of the flow panel. Section 3 reports the scale of external dependence across occupations. Section 4 turns to who moves. Section 5 estimates the gravity model. Section 6 uses departures from the gravity benchmark to locate occupational agglomeration. Section 7 tests the measures against outside data, and Section 8 sets out the limitations.

## 2 Data and construction

### Source and coverage

The flows are built from IPUMS USA extracts of the American Community Survey one-year files for 2005 to 2024 (Ruggles et al., 2024), 59.8 million person records in total. There is no 2020 vintage, because the Census Bureau suspended one-year collection in that year, so the panel runs to nineteen vintages with a gap in the middle, and any calculation crossing 2019 to 2021 measures a two-year change. Occupation is coded on the OCC2010 basis, the IPUMS harmonisation, which yields 479 occupations present throughout the panel and avoids the step changes at the 2010 and 2018 revision boundaries that a crosswalk to the contemporaneous coding introduces.

A worker is counted as an interstate mover if the state of residence one year previously is a valid state other than the current one. The rule selects a strict subset of the survey's own interstate migration flag: no record is classified as a mover by the rule and not by the flag, while 0.7 per cent of movers carry the flag without a recorded origin state and are excluded, since a flow cannot be assigned to a pair of states without one.

### **Flows and zeros**

Flows are counted for every origin, destination, occupation and year, and the zeros are retained. Of the cells in the estimation grid, 91.6 per cent record no movers, and a corridor carrying nobody in a given year is informative about the difficulty of that move. Dropping the zeros, which taking logarithms of flows requires, would estimate the model on a selected sample of large corridors and would introduce a bias whose direction cannot be signed in advance.

### **Wages**

The wage panel gives the mean log hourly wage by occupation, state and year, built from wage and salary workers who did not cross a state line in the year. The restriction is necessary because the destination wage would otherwise absorb the selection the model exists to detect: workers who move in response to a good offer earn more on arrival, and an elasticity estimated on wages that include theirs would be partly a construction of the measurement. Movers are 2.3 per cent of the workforce, so little is lost by excluding them.

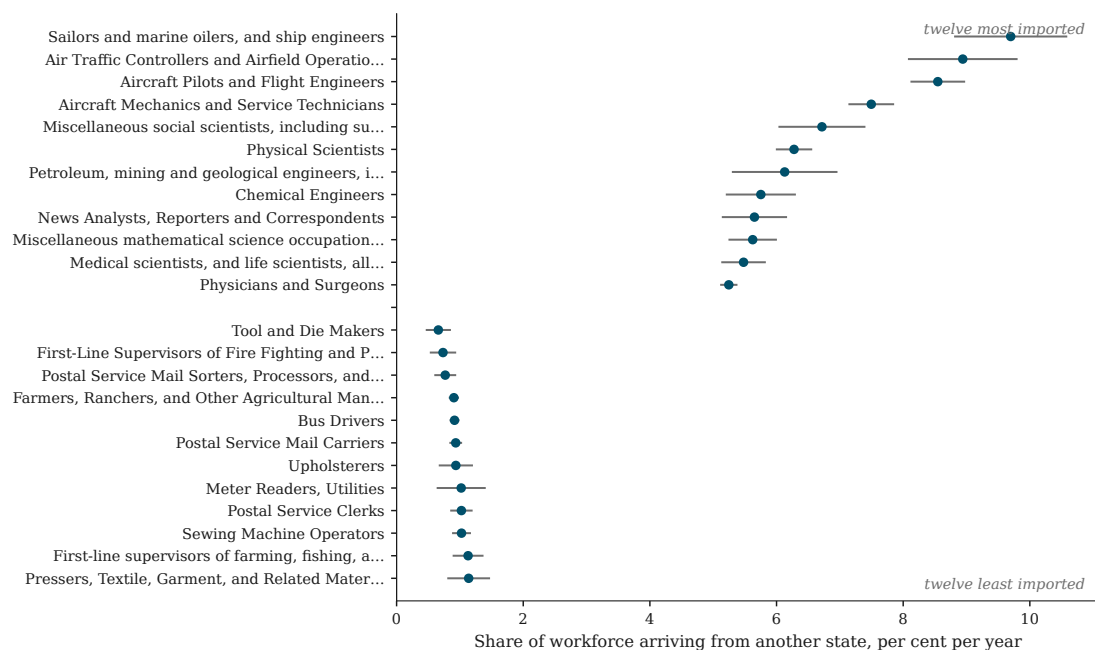
Hourly pay is computed from annual wage income, usual weekly hours and weeks worked. The change from an annual to an hourly concept does not buy precision: reliability, defined as the share of observed variation that is not sampling noise, is 0.762 for cross-sectional wage differences on either measure. It does remove an arbitrary threshold on hours that the annual measure required, and it corresponds to the rate at which a job is actually offered.

### **Distance and standard errors**

Distance is the great-circle distance between state centres of population, which places the origin of Nevada near Las Vegas where its population is concentrated and away from its sparsely inhabited geographic centre, and contiguity is derived from the Census county adjacency file. Standard errors on the sourcing measures are computed by Taylor linearisation on the survey's strata and clusters, which the ACS design supports without collapsing: there are 2,462 strata in the final year, a minimum of 191 clusters per stratum and no singletons. The median relative standard error across occupations is 6.1 per cent, and 345 of 475 occupations fall below 10 per cent.

## **3 The scale of external dependence**

External dependence is the share of a state's employed workforce in an occupation that arrived from another state within the year. Pooled across states and vintages it ranges from 0.66 per cent for tool and die makers to 9.70 per cent for sailors and marine oilers, a thirteenfold spread whose



**Figure 1:** External dependence by occupation, pooled 2005 to 2024. Points are the share of the employed workforce arriving from another state within the year; bars are 95 per cent intervals from Taylor linearisation on the survey design. Occupations with fewer than 5,000 person records are omitted.

ends remain an order of magnitude apart at the bounds of their confidence intervals, with the most imported occupation bounded below at 8.80 per cent and the least bounded above at 0.86. Figure 1 reports the two ends of the distribution.

Occupations at the top of the distribution either serve networks that span states by construction, as sailors, pilots, air traffic controllers and aircraft mechanics do, or draw on a national market for scarce training, as physical scientists, chemical engineers and physicians do. Occupations at the bottom are attached to a fixed local asset such as a farm, a postal route, a stretch of highway or a municipal fire service. Licensure organises the distribution poorly: physicians and veterinarians are among the most heavily licensed occupations in the country and sit near the top of the ranking, while postal workers require no licence and sit near the bottom.

Gross movement runs at roughly twice net movement. Nationally, arrivals plus departures come to between 4.5 and 5.4 per cent of employment against an arrival rate near 2.3 per cent, and the gap between the two widens at state level, so that the District of Columbia replaces close to a fifth of its workforce across state lines each year while its net balance is approximately zero. A measure built on net flows alone would record no activity in that case.

## 4 Migration and the life cycle

That migration is concentrated among the young is long established (Rogers and Castro, 1981). The occupational resolution adds two things to it: the pattern holds in every occupation measured, and the flow is symmetric in age between arrivals and departures.

Annual external dependence falls monotonically with age, from 4.54 per cent among workers



**Figure 2:** Left: annual external dependence by age band, all occupations pooled. Right: mean age of arrivals against mean age of the existing workforce, one point per state and occupation with at least 100 arrivals and 100 departures. Points below the diagonal indicate arrivals younger than the workforce they join.

under 25 to 0.98 per cent among those aged 55 and over, a gradient of four and a half to one shown in the left panel of Figure 2. The mean age of arrivals is 34.0 years, the mean age of leavers is 34.0, and the mean age of the workforce they join or leave is 41.3. The equality of the first two figures is not produced by averaging opposing tendencies across cells, since the median gap between arrival and departure ages across state and occupation cells is  $-0.2$  years and two thirds of cells fall within two years of each other.

Every point in the right panel of Figure 2 lies below the diagonal. Across all occupations with at least 2,000 observed arrivals the incomers are younger than the workforce they join, from 2.9 years younger for waiters to 11.9 years for aircraft mechanics, 11.0 for physicians and 10.8 for postsecondary teachers. Interstate migration therefore leaves the age composition of a state's occupational workforce roughly where it found it, exchanging early-career members for early-career members drawn from elsewhere.

A state facing a shortage of experienced workers in an occupation cannot recruit its way out of it through interstate migration, because the channel does not carry experienced workers in any occupation at any level of pay, which leaves training and early-career retention as the available instruments.

## 5 The gravity model

### Specification

Flows are estimated by Poisson pseudo-maximum likelihood, which accommodates the zeros and the heteroskedasticity that a log-linear specification mishandles (Santos Silva and Tenreiro, 2006), on 4.02 million origin by destination by occupation by year cells. Estimating one occupation at a

**Table 1:** Wage elasticity of interstate migration flows, Poisson pseudo-maximum likelihood, standard errors clustered by state pair.

| Fixed effects                             | Elasticity   | Std. error | <i>p</i> |
|-------------------------------------------|--------------|------------|----------|
| Origin, destination, year                 | 0.075        | 0.022      | 0.0006   |
| Origin-year, destination-year, occupation | 0.109        | 0.026      | <0.0001  |
| + state pair                              | 0.108        | 0.026      | <0.0001  |
| + occupation-year                         | <b>0.114</b> | 0.026      | <0.0001  |

time is not viable, and the attempt returns a null that is an artefact of the design: within a single occupation the wage differential is identified only from movement in a state by occupation cell mean over time, and a reliability calculation attributes 54 per cent of that movement to sampling noise, so the coefficient is attenuated to roughly half its size against standard errors too wide to detect it. Pooling occupations identifies the elasticity from variation across occupations within a state pair and year, whose reliability is 0.762, and it permits origin-by-year and destination-by-year effects that absorb the time-varying multilateral resistance which would otherwise sit in the error term (Anderson and van Wincoop, 2003; Bertoli and Fernández-Huertas Moraga, 2013; Head and Mayer, 2014).

### Wage and distance elasticities

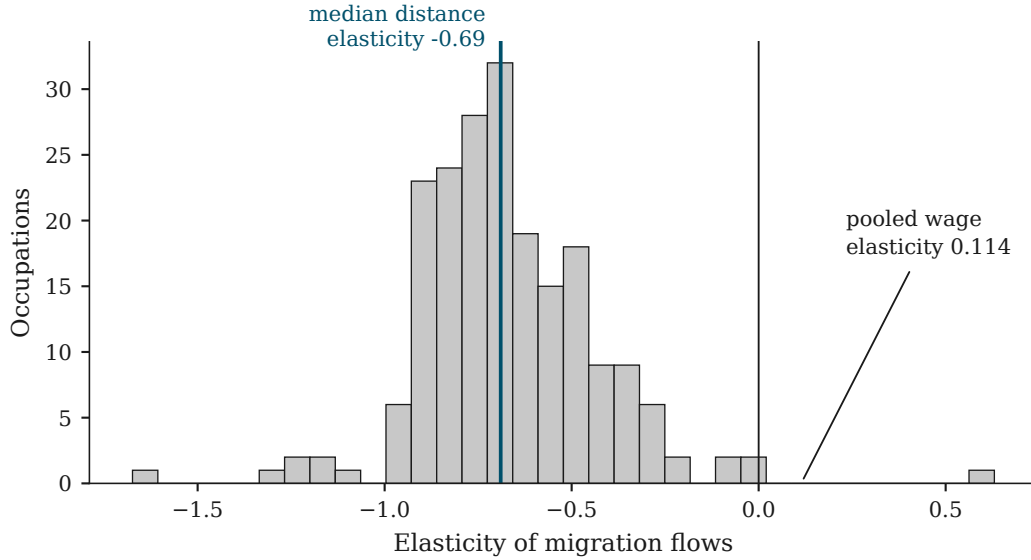
Table 1 reports the wage elasticity across four specifications of increasing severity. The estimate is stable and it rises as effects are added, which indicates that omitted multilateral resistance biased the looser specifications towards zero. The preferred estimate survives origin-by-year, destination-by-year, occupation-by-year and dyad effects together, so it is identified from the way wage differentials vary across occupations within one state pair in one year.

An elasticity of 0.114 means that a destination paying ten per cent more than the origin draws about one per cent more migration in that occupation. Inverted, a state seeking ten per cent more arrivals through pay alone would have to raise the occupation’s wage by something approaching ninety per cent. Correcting for the measurement reliability of the wage variable raises the estimate to approximately 0.15 and leaves that conclusion intact.

The distance elasticity is an order of magnitude larger in its effect, and unlike the wage elasticity it can be estimated occupation by occupation, because it is identified from variation across 2,550 state pairs and not from movement over time. It is statistically distinguishable from zero in 93 per cent of the 203 occupations with sufficient corridors, and it varies threefold across them, from  $-0.04$  for editors and  $-0.06$  for chemists to  $-1.21$  for printing press operators and  $-1.31$  for roofers. Figure 3 places the pooled wage elasticity against that distribution. A coefficient near zero belongs to an occupation recruited nationally, and one near unity to an occupation whose recruitment beyond the neighbouring states returns very little.

### Turnover and net balance

Occupations differ twentyfold in the share of their workforce crossing a state line each year in either direction, from 1.29 per cent for tool and die makers to 25.52 per cent for avionics technicians, and



**Figure 3:** Distribution of the distance elasticity across 203 occupations estimated separately, with the pooled wage elasticity marked for scale.

very little of that traffic accumulates on either side. Aircraft pilots move 17.09 per cent of their workforce across state lines annually for a net balance of 0.000 per cent, and aircraft mechanics move 14.98 per cent for 0.011. Figure 4 shows the pattern across all occupations, with net balance confined to a narrow band around zero at every level of gross movement.

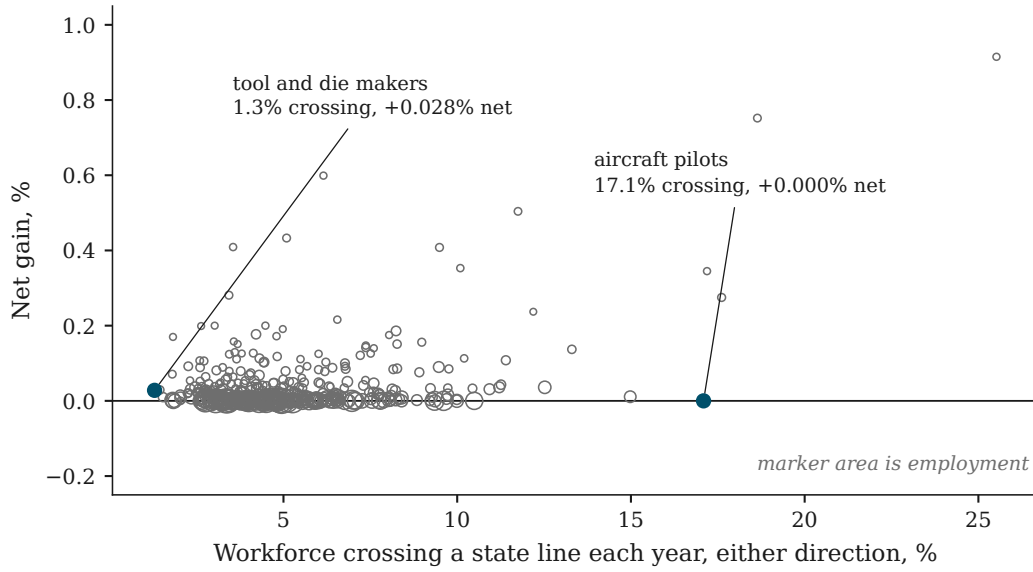
The distinction matters for the instrument a state can use, since in a high-turnover occupation each worker attracted is matched by one lost, so that additional recruitment replaces the outflow without raising the stock and retention becomes the binding constraint, while in a low-turnover occupation the channel is too narrow to recruit through at all and the workforce has to be produced locally.

## 6 Departures from the benchmark

Once distance, contiguity, wage differentials and everything constant about each state in each year have been accounted for, the model still leaves a systematic component in the state by occupation cells. Figure 5 plots observed against predicted arrivals.

The residual reproduces occupational concentrations that were not supplied to the model. Washington state draws 2.82 times its predicted intake of software developers and California 2.03, the District of Columbia 3.64 times its predicted intake of lawyers, Michigan 4.16 times its predicted intake of mechanical engineers, and New York 2.76 times its predicted intake of securities and financial services agents. Under-supply appears with equal clarity, with South Carolina, Tennessee and Kentucky drawing between 0.28 and 0.44 of their predicted intake of software developers. California draws twice its predicted software developers and half its predicted home health aides, which is the pattern a high cost of living would produce by sorting arrivals on what they earn.

Deviations are judged against Poisson sampling variation, because a ranking on the raw ratio surfaces only small-cell artefacts: the standard deviation of the ratio falls from 0.597 in cells below



**Figure 4:** Annual crossing rate against net balance, one point per occupation, marker area proportional to employment.

2,000 expected arrivals to 0.203 in cells above 50,000. On the noise-adjusted measure, 43 per cent of cells exceed a  $z$  of 2, against the 5 per cent that sampling variation alone would produce.

The concentrations are stable over the period. Aggregating the residual into four multi-year windows on a balanced panel, the correlation between the 2005 to 2009 and the 2021 to 2024 positions is 0.543, and the noise-corrected spread of the residual is flat across the four windows at 0.263, 0.276, 0.257 and 0.275. The geography of occupational concentration does not measurably move across twenty years that contain a financial crisis, a pandemic and a rapid expansion of remote work.

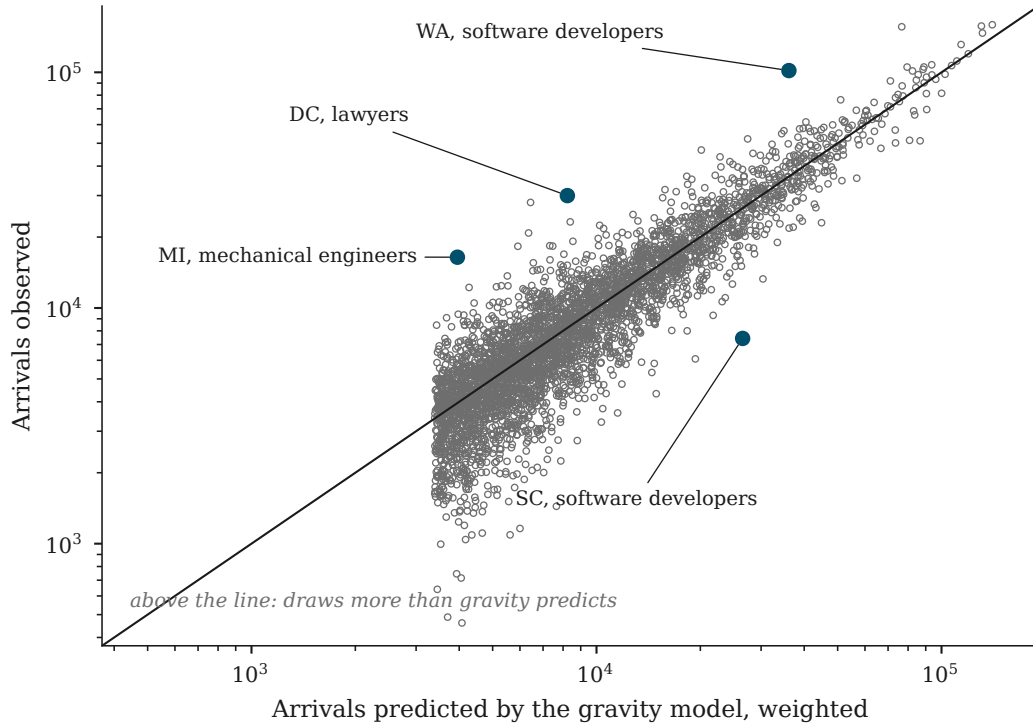
## 7 Testing the measures against outside data

Every measure above is built from household survey records. The Occupational Employment and Wage Statistics is collected from employers, so agreement between the two carries information that a comparison within a single source would not. Ten hypotheses were specified before the results were examined, all ten are reported, and  $p$  values carry a Benjamini-Hochberg correction across the battery (Benjamini and Hochberg, 1995). Regressions are weighted by employment and clustered by occupation and by state.

Three of the ten survive correction. The gravity residual tracks the location quotient with a standardised coefficient of 0.134 and an adjusted  $p$  of  $4.4 \times 10^{-7}$ , the result shown in Figure 6, which establishes that the residual carries a real signal about occupational concentration. External dependence predicts subsequent employment growth at 0.044, and net inflow predicts it at 0.029.

The wage tests return nothing. Whether the predictor is the gravity residual, external dependence or net inflow, the relationship with five-year wage growth is  $-0.0004$ ,  $-0.0014$  and  $+0.0001$  respectively, against standard errors between 0.004 and 0.007, which is enough precision to treat all three as estimated zeros. The consequence is direct: a state drawing materially fewer workers than





**Figure 5:** Observed against gravity-predicted arrivals by state and occupation, pooled 2005 to 2024, log scales. Cells below roughly thirty expected sampled movers are suppressed.

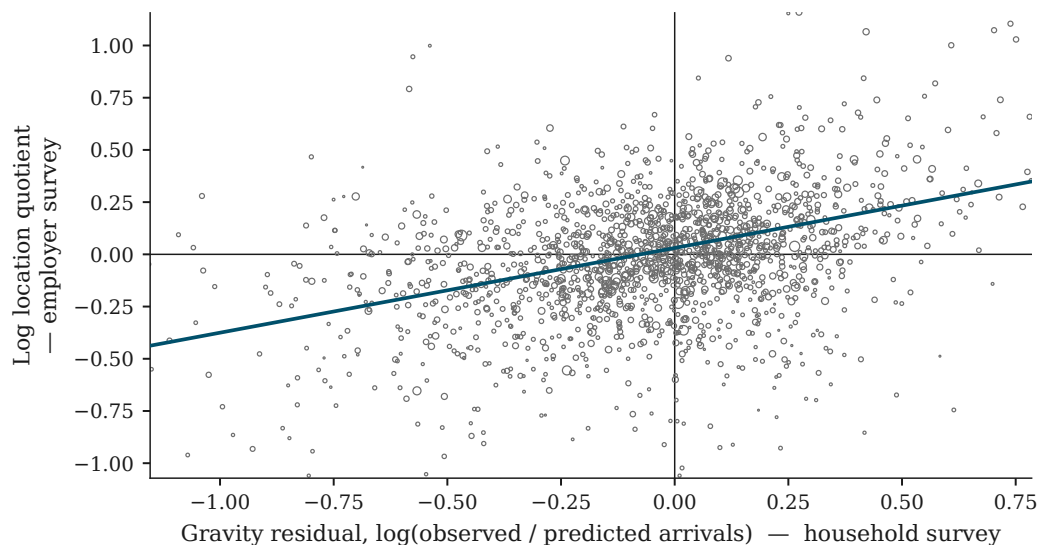
the benchmark predicts sees no faster wage growth, so the residual cannot be read as a shortage signal and is not presented as one here. A tenth result points the same way, with states paying above the national rate for an occupation importing very slightly fewer of them, wrong-signed and not significant after correction, which is the sign a cost of living effect would carry.

## 8 Limitations

Occupation and age are recorded at the time of survey, after any move has taken place. An arrival counted in an occupation either practised it before moving or entered it on arrival, and the data cannot separate the two cases. For occupations with long training requirements the ambiguity is minor, while for easily entered occupations the sourcing measure is an upper bound on imported trained workers, and the claims made here are restricted accordingly.

Coverage across state and occupation cells is uneven. The results are secure nationally, by occupation, and for large combinations of state and occupation, but only 4,026 of 24,102 state by occupation cells reach thirty arrivals and thirty departures even when all nineteen vintages are pooled, so a complete lookup across every state and every occupation is not available from these data and is not offered. Cells below the threshold are suppressed and not interpolated. The absent 2020 vintage makes any calculation spanning 2019 to 2021 a two-year change.

Nothing estimated here identifies a causal effect. The residual locates where a state diverges from its benchmark and is silent on the reason, and the association between migration and employment growth runs in a direction these data cannot determine.



**Figure 6:** The gravity residual, from household survey microdata, against the location quotient, from the establishment survey. Marker area proportional to employment; the line is the employment-weighted fit.

## 9 Conclusion

Interstate migration supplies occupational workforces through a channel that varies thirteenfold in width across occupations, that is narrow in every one of them, and that carries workers of a single age range. The measures constructed here put numbers on both facts, report them with the uncertainty the survey design implies, and agree with an independent employer survey wherever agreement is testable.

Three of the results bear on a state deciding how to staff an occupation. Pay is a weak instrument, since a ten per cent advantage buys about one per cent more arrivals and the wage tests against outside data return nothing. Recruitment cannot supply experience, since no occupation draws arrivals older than the workforce they join. In the occupations where movement is heaviest, arrivals and departures cancel, so the constraint on the stock is retention.

The deviations from the gravity benchmark are large, stable across two decades and so far unexplained, and the apparatus assembled here locates them without accounting for them. Occupational licensure compacts, which take effect between specific pairs of states on specific dates, supply variation of the kind that could account for part of them, and that is the direction the work takes next.

## References

- James E. Anderson and Eric van Wincoop. Gravity with gravitas: A solution to the border puzzle. *American Economic Review*, 93(1):170–192, 2003.
- Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate. *Journal of the Royal Statistical Society: Series B*, 57(1):289–300, 1995.

- Simone Bertoli and Jesús Fernández-Huertas Moraga. Multilateral resistance to migration. *Journal of Development Economics*, 102:79–100, 2013.
- Keith Head and Thierry Mayer. Gravity equations: Workhorse, toolkit, and cookbook. In *Handbook of International Economics*, volume 4, pages 131–195. Elsevier, 2014.
- John Kennan and James R. Walker. The effect of expected income on individual migration decisions. *Econometrica*, 79(1):211–251, 2011.
- Raven Molloy, Christopher L. Smith, and Abigail Wozniak. Internal migration in the united states. *Journal of Economic Perspectives*, 25(3):173–196, 2011.
- Andrei Rogers and Luis J. Castro. *Model Migration Schedules*. International Institute for Applied Systems Analysis, Laxenburg, 1981.
- Steven Ruggles, Sarah Flood, Matthew Sobek, et al. IPUMS USA: Version 15.0 [dataset], 2024.
- J. M. C. Santos Silva and Silvana Tenreyro. The log of gravity. *The Review of Economics and Statistics*, 88(4):641–658, 2006.